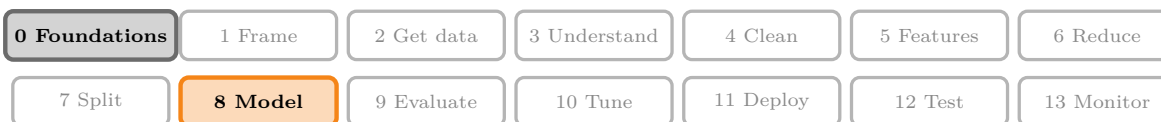


Voting Ensemble: Why Majority Voting Works

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Where this fits: Pipeline map steps 0 (Foundations), 8 (Model). See the [Course map](#).



- **Builds on:** Ensemble learning ([Note ML-009](#)); Cross-validation ([Note ML-028](#)); Independent and mutually exclusive events ([Note MA-016](#)); Probability distributions ([Note MA-003](#)).
- **Compare with:** Bagging ([Note ML-099](#)); Stacking and blending ([Note ML-121](#)); Poisson distribution ([Note MA-022](#)).

1. Overview

Key point: A voting ensemble trains several models on the same data and lets them vote. If the models are independent and each is right more than half the time, the vote is right more often than any one of them.

A **voting ensemble** (G-2096) is the simplest ensemble ([introduction to ensemble learning Note](#), section 4.1). This Note and the next two cover it:

- this Note: the core idea (section 2), a puzzle (section 3), the two assumptions voting needs (section 4), and why voting works, with probability (section 5);
- hard and soft voting, code and hyperparameters: the [voting classifier Note](#);
- the same ideas for regression: the [voting regressor Note](#).

2. The core idea

Key point: Train every model on the same data, independently; to predict, ask them all and take the majority (classification) or the mean (regression).

Take a few models, say M1, M2 and M3. The models inside an ensemble are its **base models** (G-260). They can be different algorithms or the same algorithm with different settings; the [voting classifier Note](#) shows both.

Training: each model is trained on the **same** dataset, independently of the others.

Prediction: a new query point x_q (the one observation we want an answer for, for example one new flower with petal length 4.5 cm) goes to every trained model, and the ensemble returns the **majority vote** (G-1146; the class most models predict) for classification, or the mean of their outputs for regression, as in the [ensemble learning Note](#), section 3.

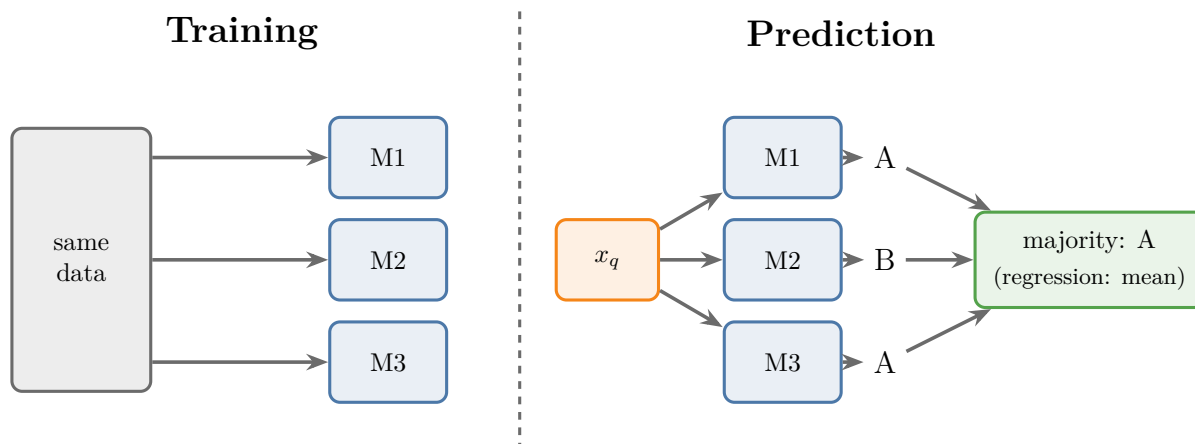


Figure 1: Training: every model learns from the same data. Prediction: every model answers the query, and the majority (or mean) is returned

In Figure 1, follow the query x_q : it reaches all three models, two answer A and one answers B, so the ensemble answers A.

Two small examples with numbers:

- **Classification.** For a query point, M1 predicts 1, M2 predicts 0 and M3 predicts 0. Class 0 has two votes of three, so the ensemble predicts 0.
- **Regression.** M1 predicts 0.5, M2 predicts 0.8 and M3 predicts 0.55. The ensemble predicts their mean, in two steps:

$$0.5 + 0.8 + 0.55 = 1.85$$

$$1.85/3 = 0.617$$

Those two steps are the whole algorithm. A voting ensemble works like an election, which is where the name comes from.

3. The puzzle

Key point: Can a vote beat its best member? With accuracies 0.7, 0.6 and 0.55, not quite (0.673); with three models of accuracy 0.7, yes (0.784).

Suppose one model has accuracy 0.7, another 0.6 and a third 0.55. Can a vote of the three beat 0.7, the best of them? How could a vote reach further than its best member?

The answer needs two assumptions and a little probability. Figure 2 shows where it leads, for independent models.

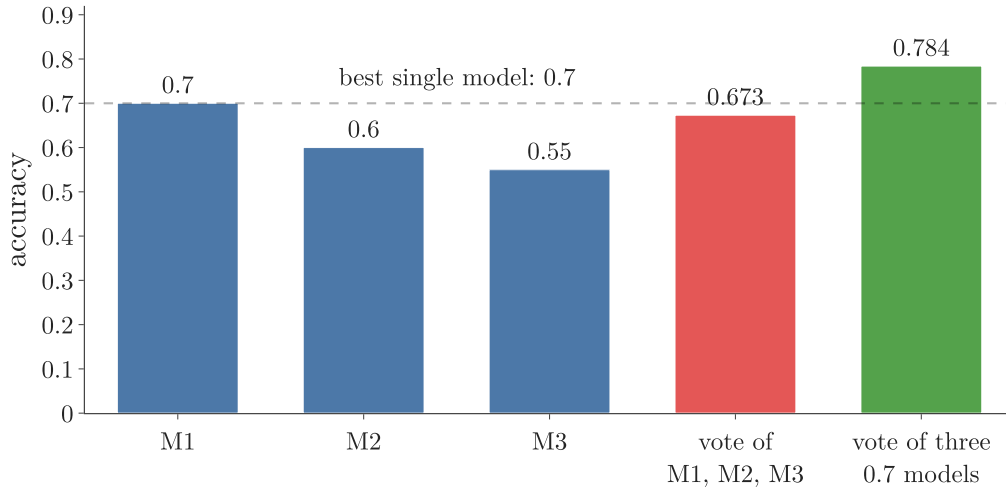


Figure 2: Accuracy of each model and of two votes of independent models: M1, M2, M3 (0.7, 0.6, 0.55) and three models of accuracy 0.7

In Figure 2, compare the red and green bars. The red bar is the vote of M1, M2 and M3; the green bar is a vote of three models that are each 0.7. The vote is right when at least two models are right. Each model is right with probability 0.7, 0.6 and 0.55, so it is wrong with probability 0.3, 0.4 and 0.45. Because the models are independent (section 4), the probability of one combination is the product of its three numbers. Four combinations have at least two models right:

M1	M2	M3	Product	Probability
right	right	right	$0.7 \times 0.6 \times 0.55$	0.231
right	right	wrong	$0.7 \times 0.6 \times 0.45$	0.189
right	wrong	right	$0.7 \times 0.4 \times 0.55$	0.154
wrong	right	right	$0.3 \times 0.6 \times 0.55$	0.099

Adding the last column, one step at a time:

$$0.231 + 0.189 = 0.420$$

$$0.420 + 0.154 = 0.574$$

$$0.574 + 0.099 = 0.673$$

The vote is right with probability 0.673, just below 0.7, because the two weaker models can outvote the best one. Three equally good models of 0.7 reach 0.784 (section 5).

4. The two assumptions

Key point: (1) The models must be independent: the more different, the better. (2) Each model must be right more than 50% of the time.

Assumption 1: the base models are independent. In probability terms the models' mistakes are **independent events** (G-934): knowing one model is wrong tells us nothing about the others. In plain words: when one model is wrong, the others should not be wrong for the same reason. The more different the models are, the better the vote works, just as a mixed quiz-show audience covers more topics than one of programmers only (the [ensemble learning Note](#), section 3.1); if they are very similar, voting gains little.

Assumption 2: every model's accuracy is above 50%. A model that is right less than half the time does harm: a vote of such models is **worse than the worst of them**.

For two classes, 50% accuracy is what random guessing gives. Beating a coin toss is not hard, so in practice voting is a safe technique that usually helps.

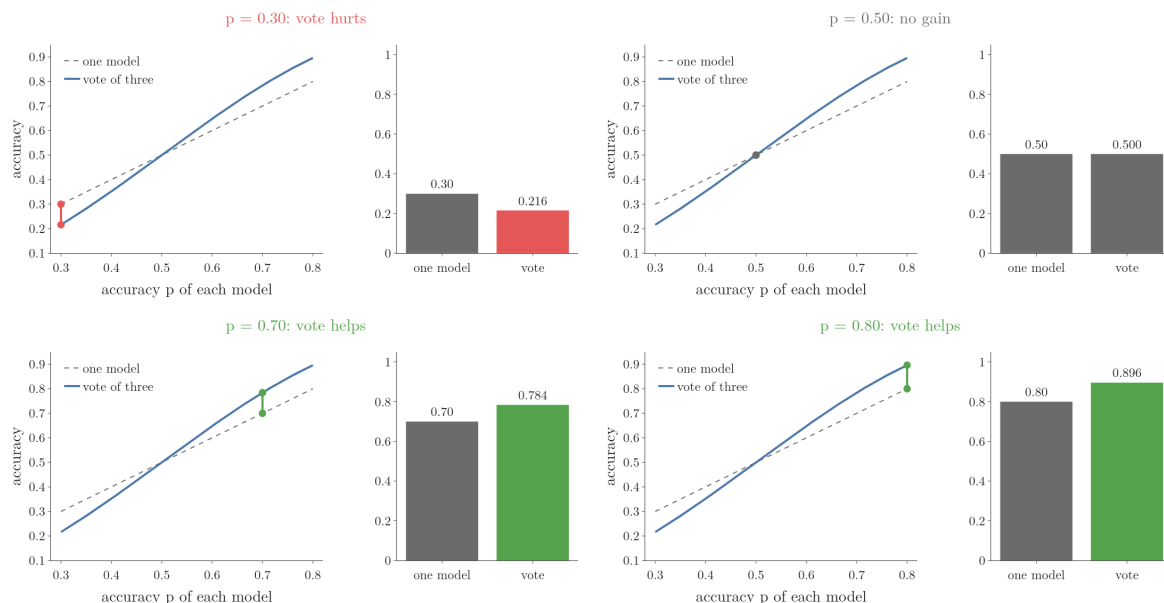


Figure 3: Three independent models, each right with probability p , as p sweeps from 0.30 to 0.80: the vote's accuracy against one model's

In Figure 3, watch the blue curve cross the dashed line at $p = 0.5$: below it the vote is worse than one model, above it better, and the gap grows as p moves away from 0.5.

5. Why voting works: the probability

Key point: With three independent models of accuracy 0.7, the vote is right whenever at least two are right, which happens with probability 0.784.

Take three independent models, M1, M2 and M3, each with accuracy 0.7. On a new point, each is right with probability 0.7 and wrong with probability 0.3.

Because the models are independent, the probability of a combination of outcomes is the product of the separate probabilities (the multiplication rule, [independent events Note](#)). Figure 4 lists all 8 combinations.

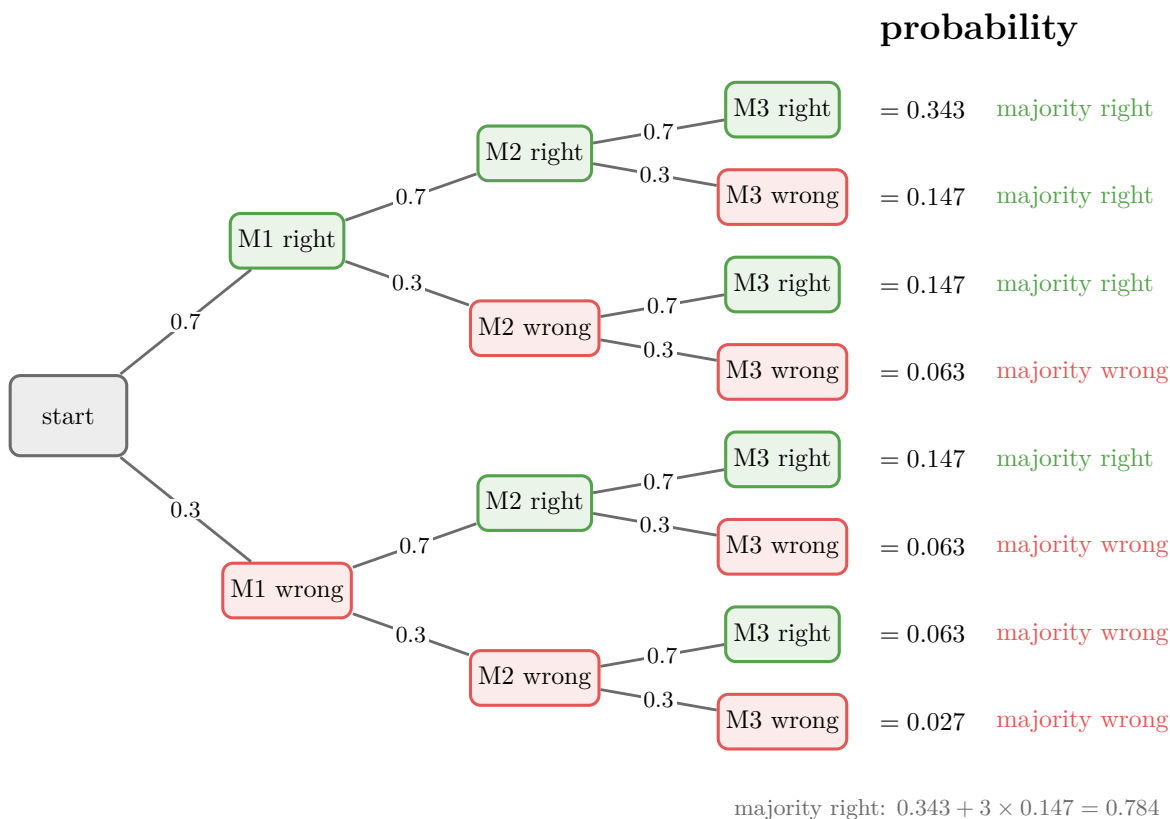


Figure 4: Every combination of right and wrong for three independent models of accuracy 0.7

The majority is right when **at least two** models are right. Four of the eight combinations qualify.

Figure 5 adds those four up, one at a time:

1. All three right:

$$0.7 \times 0.7 \times 0.7 = 0.343$$

2. M1 and M2 right, M3 wrong:

$$0.7 \times 0.7 \times 0.3 = 0.147$$

Running total 0.490.

3. M1 and M3 right, M2 wrong: 0.147. Running total 0.637.

4. M2 and M3 right, M1 wrong: 0.147. Running total 0.784.

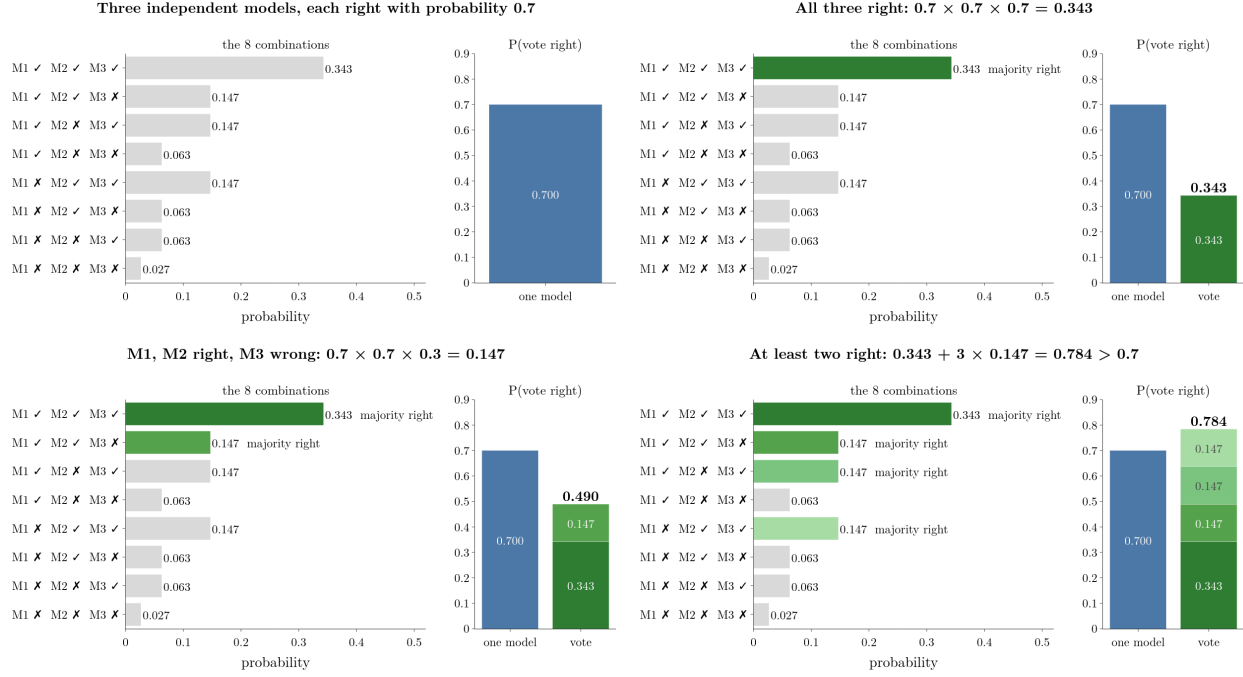


Figure 5: The eight combinations of Figure 4 as bars. The four where the majority is right light up in turn, and their probabilities stack up on the right: 0.343, 0.490, 0.637, 0.784, above the 0.7 of one model

In Figure 5, watch the stacked bar on the right pass the blue bar of a single model when the last piece is added. The general rule behind the four steps:

In words: add the probabilities of “all three right” and of the three ways to have exactly two right.

Formula: let p be the probability that one model is right ($p = 0.7$ above). The first term, p^3 , is “all three right”. The second term counts the three ways to have exactly two right (M3 wrong, M2 wrong or M1 wrong); each way has probability $p^2(1 - p)$, so the three ways give $3p^2(1 - p)$. With $p = 0.7$ the two pieces are:

$$p^3 = 0.7^3 = 0.343$$

$$p^2(1 - p) = 0.7^2 \times 0.3 = 0.147$$

The formula is:

$$P(\text{vote right}) = p^3 + 3p^2(1 - p)$$

Example: with $p = 0.7$, in three lines:

$$0.7^3 = 0.343$$

$$3 \times 0.7^2 \times 0.3 = 3 \times 0.147 = 0.441$$

$$P(\text{vote right}) = 0.343 + 0.441 = 0.784$$

This matches the running total of the four steps above.

So three models of accuracy 0.7 make a vote of accuracy **0.784**, better than each one. The vote is wrong only when two or three models are wrong at the same time, and with independent models that is rarer than one model being wrong.

5.1 Below 50%: the vote is worse than the worst

Key point: With three models of accuracy 0.3, the vote is right with probability only 0.216.

Now let each model be right with probability $p = 0.3$. The same formula gives:

$$0.3^3 = 0.027$$

$$3 \times 0.3^2 \times 0.7 = 3 \times 0.063 = 0.189$$

$$P(\text{vote right}) = 0.027 + 0.189 = 0.216$$

A vote accuracy of about 22% is worse than every single model (30%). The value 0.216 is also $1 - 0.784$: the four combinations left over in Figure 4, with the roles of right and wrong swapped. This calculation proves assumption 2: voting amplifies whatever the models are, good or bad.

Extra: With n models (an odd number, so there are no ties), the vote is right when more than half are right. The number of right models follows a **binomial distribution**

(G-308), so

$$P(\text{vote right}) = \sum_{k > n/2} \binom{n}{k} p^k (1-p)^{n-k}$$

Reading the symbols with $n = 3$ and $p = 0.7$: $\sum_{k > n/2}$ means “add one term for every k above $n/2 = 1.5$ ”, so $k = 2$ and $k = 3$. $\binom{n}{k}$ counts the ways to choose which k models are right: $\binom{3}{2} = 3$ and $\binom{3}{3} = 1$. The two terms are

$$k = 2 : 3 \times 0.7^2 \times 0.3 = 0.441$$

$$k = 3 : 1 \times 0.7^3 \times 0.3^0 = 0.343$$

and their sum, 0.784, is the $p^3 + 3p^2(1-p)$ above. This result is known as **Condorcet’s jury theorem** (G-445; Condorcet, 1785).

5.2 More models, and correlated models

Key point: With independent models above 50%, more models push the vote toward 100%; once the models start copying each other, the benefit disappears.

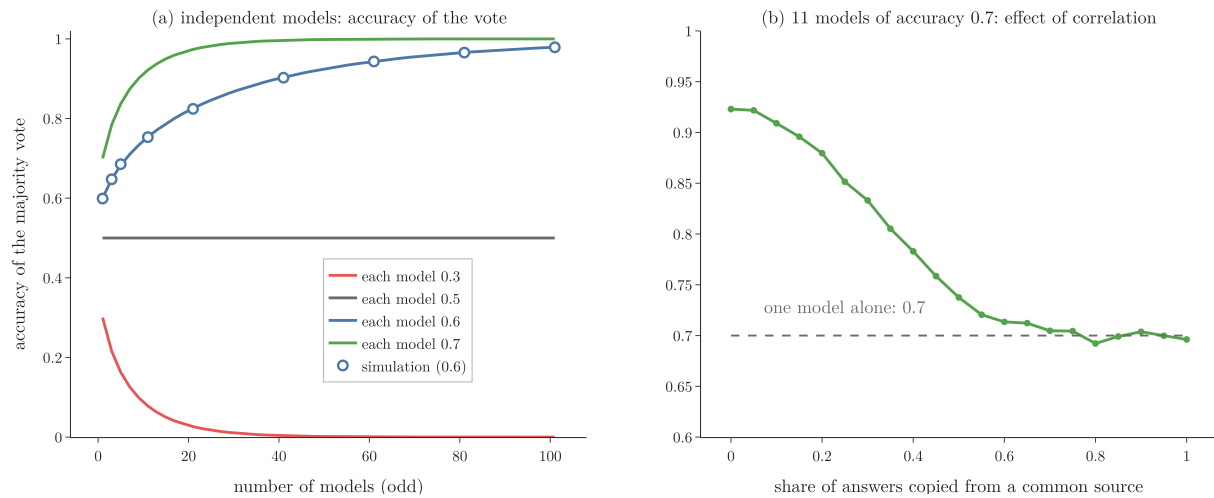


Figure 6: (a) Accuracy of the vote against the number of independent models; (b) 11 models of accuracy 0.7 whose answers are partly copied from a common source

Figure 6a uses the formula of the Extra box for 1 to 101 models:

Each model	3 models	11 models	101 models
0.7	0.784	0.922	1.000
0.6	0.648	0.753	0.979
0.5	0.500	0.500	0.500

Each model	3 models	11 models	101 models
0.3	0.216	0.078	0.000

Even models only slightly better than chance (0.6) reach a vote of 0.98 with 101 of them. Models at exactly 0.5 gain nothing, and models below 0.5 drive the vote to 0.

The dots in Figure 6a come from a simulation: 20,000 random trials of independent models of accuracy 0.6. They sit on the formula's line, so the formula is right.

Figure 6b tests assumption 1. Eleven models of accuracy 0.7 vote, but each answer is, with some probability, copied from one shared source instead of being the model's own:

- with no copying (independent models), the vote scores about **0.92**;
- with half the answers copied, about **0.74**;
- with everything copied, the eleven models act as one, and the vote scores **0.7**, no better than a single model.

Extra: Models trained on the same data are usually correlated, so real gains are smaller than Figure 6a promises: the correlation between the models limits what combining them can gain (ESL §15.2), just as Figure 6b shows. An ensemble beats its members only when they are both accurate and *diverse*, meaning they make different errors (Dietterich, 2000). Correlation is the reason ensembles work to make their models different: different algorithms (voting), different samples of the data (bagging), or both.

6. Summary

Situation	Each model right	Vote of 3 models right
Independent, better than chance	0.7	0.784 (better than each)
Independent, worse than chance	0.3	0.216 (worse than each)
Fully correlated	0.7	0.7 (no gain)

- A voting ensemble trains several models on the same data; classification takes the majority vote, regression the mean.
- It needs two assumptions: independent (different) models, and every model better than 50%.
- For three models, the chance that the vote is right is 0.784 for $p = 0.7$:

$$P(\text{vote right}) = p^3 + 3p^2(1 - p)$$

- More independent models help more; correlated models help less.

7. Sources

Built from

- CampusX, “Voting Ensemble | Introduction and Core Idea | Part 1”, YouTube, https://www.youtube.com/watch?v=_W1i-c_6rOk

Other references

- Condorcet, Marquis de (1785). *Essai sur l'application de l'analyse à la probabilité des décisions rendues à la pluralité des voix*. Paris.
- Dietterich, T. G. (2000). “Ensemble Methods in Machine Learning”. *Multiple Classifier Systems* (MCS 2000), LNCS 1857, Springer, section 1.
- Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning* (ESL), 2nd ed. Springer, section 15.2.

8. Key terms

Term	Meaning
Voting ensemble (G-2096)	Several models trained on the same data, combined by majority vote (classification) or mean (regression)
Base model (G-260)	One of the models inside an ensemble
Independent models	Models whose mistakes are unrelated, so one being wrong says nothing about the others

Term	Meaning
Binomial distribution (G-308)	The distribution of the number of successes in n independent trials with the same success probability
Condorcet's jury theorem (G-445)	A majority of independent voters, each right with probability above 0.5, is right more often than any one voter, and more so as voters are added